

# GAZE: Game-Adaptive Zone of Engagement

## Notes:

- [ ] Record and embed timeframe in method & setup

## Diagram ideas:

Input : face detection( emotions ) -> open AI create game ( based on emotions) -> robot speaks and interact ( verbal , hand and face gestures.)

- if frustrated -> hints.
- If angry -> easier game
- If happy-> celebrate and smile and give harder games
- If Bored -> speeds up
- If seem upset -> cheer them up
- ...

## Robot idea:

- [ ] integrate ws-10 for facial recognition and emotion detection
  - [ ] Generates games with openai on the fly
  - [ ] Facial recognition (Happy? frustrated?) if frustrated give hints if happy celebrate via /ws-10
  - [ ] If bored speeds it up
  - [ ] If angry - no worries lets try another game ( easier)
  - [ ] If answers are correct, it gives harder puzzles.
  - [ ] Make easier if too hard
  - [ ] If sad games to cheer him up
  - [ ] Camera - detects face ( emotions) - game logic - open ai generates response - robot speaks and reacts.
  - [ ] Robot personality - funny , sarcastic , serious.
  - [ ] Games - puzzles, riddles, trivia, memory games, word games, math games, etc.
  - [ ] Use openai to generate games and responses based on the user's emotions and performance.
  - [ ] timer to track how long the user has been playing and adjust the game difficulty accordingly.
  - [ ] scoring system to track the user's progress and provide feedback on their performance.
  - [ ] reward system to encourage the user to keep playing and improving their skills.
  - [ ] feedback system to allow the user to provide feedback on the games to change robot's considerations and personality.
  - [ ] feature to allow the user to customise the robot's personality and game preferences.
  - [ ] feature to allow the user to save their progress and continue playing later.
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# Task Allocation

## report-related:

- **Salman:** Introduction (10% of the task mark)
- **Alfie:** Background (10% of the task mark)
- **Alfie:** Method and setup (35% of the task mark)
- **Salman:** Results/outcome/or system analysis (30% of the task mark)
- **Alfie: Conclusion** (10% of the task mark)
- **Anyone:** References (5% of the task mark)

## Code-related:

Salman: making prompts

Salman: gaming logic

Alfie: facial recognition

Alfie: general code architecture

## Alfie

- Code
- Talk about results
- Write conclusion

## Salman

- Plan up
- Testing
- Small tasks
- Write introduction

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## Task (4):

Introduction (S) (<220-word)

...

Background (A) (<220-word)

...

Method & Setup (A & S) (<810-word) (

...

Results/outcome/or system analysis (S & A) (<730-word)

...

Conclusion (A) (<220-word)

...

References

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Ai Declaration stuff:

...

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**Plan 2: story-telling robot**

- Bored - adds action
- Scared - toned
- Happy - continues

**Plan 3: social-skills trainer:**

- Detects eye contact
- Detect emotions
- Detect gestures (closed off?)
- Gives feedback based on the above things